

Problem Statement

Design a **Hospital Patient Records Analysis System** using Python and Pandas. The system should analyze hospital records, identify missing data, clean the dataset and generate useful statistics.

Enter the following dataset manually

Dataset Details

Patient DataFrame (5 records with predefined missing values):

Patient	Age	Blood_Group	Disease	Bill
Alice	30	A	Flu	5000
Bob	45	B	Diabetes	6500
Charlie		O		5500
David	40		Fever	7000
Eve	50	B	Cold	4800

Empty cells in the Excel table represent NaN (missing values).

The system should:

- Store Patient, Age, Blood_Group, Disease, and Bill in a DataFrame
- Identify and display missing value counts per column
- Fill missing Age with the column mean using `fillna(mean)`
- Fill missing Blood_Group with the most frequent value using `fillna(mode()[0])`
- Remove records with missing Disease using `dropna()`
- Compute average bill amount from the cleaned dataset
- Display patients whose bill is above the average

Implement the following:

- Use Pandas DataFrame
- Use missing value handling methods
- Use filtering and aggregation
- Use data cleaning techniques
- Use descriptive statistics

Input format :

No user input required. The program uses a fixed dataset of 5 patient records with predefined missing values.

Output format :

The first line of output prints **Missing Values:** followed by the count of missing values per column.

The second line of output prints **Cleaned Dataset:** followed by the cleaned DataFrame.

The third line of output prints **Average Bill:** followed by the average bill amount.

The fourth line of output prints **Patients Above Average Bill:** followed by the names of patients whose bill is above average.

Refer to the sample output for formatting specifications.

Code constraints :

Fixed dataset of exactly 5 patient records

Missing values are predefined — Age (1), Blood_Group (1), Disease (1)

After dropping missing Disease row, cleaned dataset has **4 records**

Average bill is computed on the **cleaned dataset only**

Sample test cases :

Input 1 :

Output 1 :

Missing Values:

Patient	0
Age	1

```
Blood_Group    1
Disease         1
Bill           0
dtype: int64
```

Cleaned Dataset:

	Patient	Age	Blood_Group	Disease	Bill
0	Alice	30.0	A	Flu	5000
1	Bob	45.0	B	Diabetes	6500
3	David	40.0	B	Fever	7000
4	Eve	50.0	B	Cold	4800

Average Bill: 5825.0

Patients Above Average Bill:

```
1      Bob
3      David
```

Name: Patient, dtype: object

Note :

The program will be evaluated only after the "Submit Code" is clicked.

Extra spaces and new line characters in the program output will result in the failure of the test case.